# OS ORANGE ASSIGNMENT

SRN: PES2UG24CS083

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Github link: https://github.com/armaanyt07-jpg/PES2UG24CS083-pes-vcs.git

Screenshots:

Phase 1

1 A) Output of `./test_objects`

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS083$ ./test_objects
Stored blob with hash: d58213f5dbe06299b5c2fa28e5c7d4213ea092ed0221b6k19aaafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea0922edb9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1 B) Output of `find .pes/objects -type f`

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS083-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e111ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aaffc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2cdaf5aaf2e650eda471e566fcf
```

Phase 2

2 A) Output of `./test_tree`

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS03$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2 B) Raw binary of tree object

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS083$ xxd .pes/objects/2c/
f8d83dbd3576998025de576db9ebb4 | head -20
þerctc 6677 2088 6c6f 0a
00000000: 626c 6f62 2036 086c 656c 6f0a
```

Phase 3

3 A) pes init -> pes add -> pes status

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS08$ ./pes init
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS08$ ./pes add file1.txt  file2.txt
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS08$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  test_objects
  untracked:  test_objects.c
  untracked:  Makefile
  untracked:  test_tree
  untracked:  object.c
  untracked:  index.c
  untracked:  README.md
```

3 B) cat .pes/index output

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS083$ cat .pes/index
100644 2cf8d83d9ee29543b34a87724721fdecb7e9f3a18Sd33769aDab file1.txt
100644 e00c50e16a2df38fd60bf809e181ad0284da6e6719f35f56f7f6 file2.txt
100644 e00c50e16a2df38fd60bf809e181ad0284da6e6719f35f56f7f6 file2.txt
```

Phase 4

4 A) pes log output

```
./pes log
Initialized empty PES repository in .pes/
Committed: a5f64348f3ea... Initial commit
Committed: 8dd1ea26c1bf... Add world
Committed: d95f394a2f2b... Add farewell
commit d95f394a2f2be9f3a2c288263113995c839b3301eee916bcb2377fff6e37abae
Author: Armaan <PES2UG24CS083>
Date:   1776438783

    Add farewell

commit 8dd1ea26c1bfd92411e46009248526559c33e553a63de514ad05b57d09f693cfc
Author: Armaan < Armaan <PES2UG24CS083>
Date:   1776438783

    Add world

commit a5f64348f388a6232510e7cb006507e5eebfbbca64b5a26cfbddd11ec04565
Author: Armaan < Armaan <PES2UG24CS083>
Date:   1776438782
```

4 B) `find .pes -type f | sort` showing object growth

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS08$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748bc6788d1b5caf72469754e6e6800c265ca21bd6c05
.pes/objects/27/b374421936320a8c527322a292a8f9eb9badf9be809f6137b8d32709b
.pes/objects/2c/f8d83d9ee29543b34a87724f21fdecb7e1f3a18b3d337639025de576d
.pes/objects/3f/50050781db8965d5fafbdbbbbbd32546677626ba2058ae61691fae14a
.pes/objects/53/623c175c06551b7355604f41a416c3bf7f4967a178c9f8e60018b4e9a
.pes/objects/66/2246631e2dbf46f9dee9e2c76655145a02a3830a1a52d508fafaa08f6
.pes/objects/74/4a6712449feaac827b4a71a95f48asadaa007331761c5761c94a10e61
.pes/objects/86/81e626c1bfd92411e460092485265b599c3e553a63de514a5b57cd09f
.pes/objects/96/c0d5c28ddb71906c367240d166aeffbc73b3d2574de61078825507669
.pes/objects/a5/a5f6438f3ea623265110e7cbd60507e6eebfbbca64b5a26cfbddd11ec
.pes/objects/c9/6f2fc25394579be522d5cf866c3d78ec702ff13de17f8d86b4440fbcd
.pes/objects/d5/d95f34aaf2bee9f3a2c288263113595c889b3301eee916bcb37fff6aa
.pes/objects/e00/e00c50e162df387d60bf809e181ad0234da6e6719f35f9fc658cf606
.pes/objects/7b/7ed66bcb4a702668f0d1191680806d781526badf844433880e13ad865
.pes/objects/f1/0239b620372163c2ad0606ac338119860c55edf1c56015990a1cf44e0
.pes/refs/heads/main
```

4 C) `cat .pes/refs/heads/main` and `cat .pes/HEAD`

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS083$ cat .pes/refs/heads/main
d95f394a2f2be9f3a2c288263113995c839b3301eee916bcb2377fff6e37abae
```

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS083$ cat .pes/HEAD
ref: refs/heads/main
```

Final Full integration test

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS08$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--= Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:      file.txt
  staged:      hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
```

```
armaan@DESKTOP-MBRTPOS:~/PES2UG24CS083
--- First Commit ---
Committed: 24bd3c98ae29... Initial commit

Log after first commit:
commit 24bd3c98ae29f22b267e4b271e4b2719aadc2d3bfb46a40db11514ba4fd10769ff4987
Author: Armaan <PES2UG24CS083>
Date:   1776439101

--- Second Commit ---
Committed: b596be512c83... Update file.txt

--- Third Commit ---
Committed: 6517801db05b... Add farewell

--- Full History ---
commit 6517801db05bd565da53c13744ebf27176f38994ddd4d0e26e801d49d0d2e356f
Author: Armaan <PES2UG24CS083>
Date:   1776439102

--- Add farewell
```

```
--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/24/bd3c98ae29f22b267e4b2719aadc2d3bfb46a40db11514ba4fd10769ff4987
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/65/17801db05bd565da53c13744ebf27176f38994dd4d0e26e801d49d0d2e356f
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/b5/96be512c837ce31a31a63d48948fbf4f89f745613ece39140a6dc6fb3c8ef5
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
```

Phase 5 and 6

Analysis question answers

Q 5.1

Update .pes pointers by verifying the branch file exists at .pes/refs/heads/, reading the target commit hash, and rewriting .pes/HEAD to ref: refs/heads/ using an atomic temp + rename. Resolve the target snapshot by reading the commit object, its root tree, and recursively expanding tree objects to build the full path → blob mapping.

Update the working directory by writing each blob to its file with correct mode, removing tracked files not present in the target tree, and recreating directories as needed. Rebuild the index by setting .pes/index entries to match the target tree with updated stat metadata.

This is complex because it must safely handle deletions, nested directories, permissions, partial failures, and avoid data loss when uncommitted changes overlap with target changes.

Q 5.2

Use a 3-way comparison per tracked file with base as the current HEAD tree blob hash, staged intent as the hash in .pes/index, and working file as stat plus recomputed blob hash. Build the target tree mapping and for each path differing between current HEAD and target, check if the working directory is dirty relative to the current committed state.

If the file is dirty and the target also changes that path, refuse checkout. Also detect untracked file overwrite cases where an untracked working file would be overwritten by the target tree and refuse.

This enforces the rule that checkout should not overwrite local non-committed content.

Q 5.3

In detached HEAD, .pes/HEAD stores a commit hash directly instead of a branch ref, and new commits still work by pointing to the previous detached commit while HEAD moves forward. The risk is that no branch references these commits, making them potentially unreachable.

Recovery involves creating a branch ref pointing to that commit chain or switching to a branch after anchoring the detached commit. If not recovered and no refs point to them, future GC may delete them.

Q 6.1

Use a mark-and-sweep approach where the mark phase initializes a hash set, seeds it with all branch tips, and performs DFS or BFS traversal marking commits, trees, and blobs while enqueueing referenced objects. The sweep phase scans all files in .pes/objects and deletes those whose hashes are not in the reachable set, optionally cleaning empty directories.

A hash set ensures O(1) expected membership checks and deduplication during traversal. In scale, traversal visits unique reachable objects, typically hundreds of thousands to a few million, making the complexity proportional to reachable objects plus total objects on disk.

Q 6.2

A race occurs when commit writes blob and tree objects, then commit object, and finally updates the branch ref while GC runs before the ref update and cannot see the new chain. GC treats new objects as unreachable and deletes them, after which the branch points to a broken history with missing objects.

Real systems avoid this through synchronization and locking around ref updates and GC, reachability protections like reflogs and grace periods, avoiding immediate pruning of recent objects, and ensuring objects are durable before references move.